STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



OPERATIONS

VALUE STREAM MAPPING: WHERE THE TIME GOES

Toyota's flow diagrams, a day of waiting in an oncology day hospital, current and future state, maps that stop halfway, three numbers and a ratio, and a card for your first map.

38% of a patient's day at an oncology day hospital in Spain was value-added time, at the median. Most of the rest was waiting.

VIDAL-CARRERAS ET AL., 2022

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NO. 56 · OPERATIONS

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Work is usually managed department by department, and each department can look efficient. The time the customer waits sits between them. Value stream mapping draws one flow from end to end, shows how much of its time is work, and sketches the next state before anything changes.

Mike Rother and John Shook took a simple Toyota diagram and made it a method. In an oncology day hospital in Spain, a map showed patients spending most of the day waiting on a laboratory and a pharmacy whose work took minutes. A review of 80 healthcare publications found that many maps stop at the current state. Karen Martin and Mike Osterling measure a flow with three numbers and a ratio, and the card at the end is for your first map on paper.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Learning to see

In 1998 Mike Rother and John Shook wrote that a simple tool had been right under their noses. At Toyota it was called material and information flow mapping, and it was used to draw the current and future state of a process. Rother noticed it while studying Toyota; Shook had known it for over ten years without thinking it important.

“The challenge lies in seeing it.”

Wherever there is a product for a customer, they write, there is a value stream. Dan Jones and Jim Womack describe the difficulty from their own walks through plants: most managers want to stand in one place and look at their machine, their department, their plant. These often score well on the usual measures. When the managers follow the product instead, the stream as a whole turns out to perform poorly.

OUR READING

A department can hit its target while the customer waits. The map swaps the question of how busy each step is for how long the product sits between steps.

Sources: Mike Rother & John Shook, Lean Enterprise Institute, 1999 (via Lean Enterprise Institute, introduction and Part I excerpts); Dan Jones & Jim Womack, Lean Enterprise Institute, 2011 (via Lean Enterprise Institute, introduction excerpt)

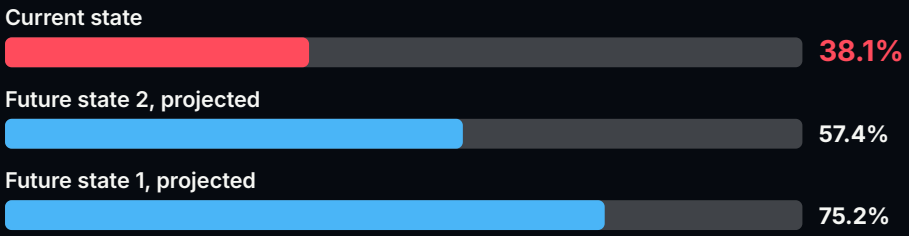
From the authors' introduction of May 1998; the workbook appeared at the Lean Enterprise Institute in 1999. The quote is confirmed only in secondary sources. Jones and Womack: Seeing the Whole Value Stream (2011).

THE NUMBERS

A day of waiting

Pilar Vidal-Carreras and colleagues mapped the day of adults receiving chemotherapy at the oncology day hospital of a large public hospital in Spain: a blood test, a visit to the oncologist, then the treatment. Patients waited 300 minutes; the median stay was 446.

VALUE-ADDED TIME AS A SHARE OF THE STAY, MEDIAN



The laboratory could test a sample in under 15 minutes, yet the next appointment came an hour later. A dose took 15 minutes to prepare on average, yet the pharmacy would not promise it in under three hours: requests from the whole hospital arrived at once, with no priority.

OUR READING

The work was quick. What made the day long was the time between departments, which no department answered for.

Source: Pilar I. Vidal-Carreras, Julio J. Garcia-Sabater & Juan A. Marin-Garcia, International Journal of Environmental Research and Public Health 19(7), 2022

The future states are designs that have not been put in place, so their gains are projected rather than measured. One hospital; the number of patients observed is not given.

THE MODEL

Current state, future state

Rother and Shook map one product family from door to door. They trace its path from the customer back to the supplier and draw two flows: the material, and the information that tells each step what to do next.

01	02	03	04
Product family	**Current state**	**Future state**	**Plan**
Chosen from the customer end: what, how much, how often?	Drawn on the floor, walking the flow.	How value should flow. The most important map.	One page on how to get there, reviewed regularly.

The arrows run both ways: ideas for the future come up while the present is being mapped. In their words, “a current state without a future state is not much use.” One person with authority across departments, a value-stream manager, leads the work. Improving the flow is “management doing kaizen”, while teams improve their own steps.

OUR READING

On a map, most boxes look busy. The lines between them are what make it worth drawing.

Source: Mike Rother & John Shook, Lean Enterprise Institute, 1999 (via Lean Enterprise Institute, introduction and Part I excerpts)

Steps and quotes follow Part I of Learning to See, as excerpted by the Lean Enterprise Institute. The commentary is ours.

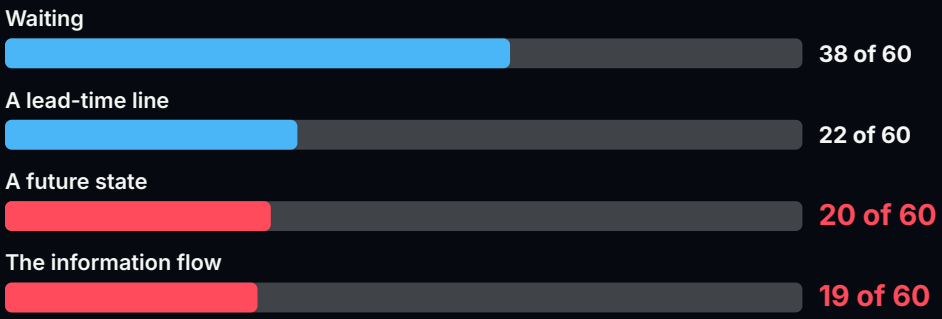
More in the essay Where part of the order goes missing

WHAT THE RESEARCH SAYS

Maps without a future

Juan Marin-Garcia and colleagues reviewed 80 publications on value stream mapping in healthcare from 2015 to 2019. Twenty talked about a map but drew none. For the other 60, the team checked what the map showed.

WHAT THE 60 DRAWN MAPS SHOW



No study reported a negative or barely noticeable result, and the authors ask whether publication bias plays a part. After reviewing 131 articles from five sectors (1999–2016), Wenchi Shou and colleagues conclude that what matters is what value and waste mean in each kind of flow, and whether the classic lean metrics fit it.

OUR READING

Which line on your current map would you shorten first? Until that is decided, the map is only a photograph.

Sources: Juan A. Marin-Garcia, Pilar I. Vidal-Carreras & Julio J. Garcia-Sabater, International Journal of Environmental Research and Public Health 18(3), 2021; Wenchi Shou, Jun Wang, Peng Wu, Xiangyu Wang & Heap-Yih Chong, International Journal of Production Research 55(13), 2017 (via Abstract at Semantic Scholar and Western Sydney University)

Waiting and the lead-time line are read off a figure; the other two follow the text (the figure shows 21 and 18). Shou's sectors: manufacturing, health care, construction, product development, services.

HOW IT IS MEASURED

Three numbers and a ratio

Karen Martin and Mike Osterling map office and service work. For every step on the map they note two times and one measure of quality; a timeline under the steps adds them up.

01

Process time

the time it takes to actually do the work

02

Lead time

from the moment work is available to a step until it is passed on

03

%C&A

percent complete and accurate: how often the next step can use the work without correcting, adding or clarifying

Work that arrives at 1 pm and leaves at 3 pm has a lead time of two hours, even if it takes 20 minutes. The activity ratio is the sum of the process times divided by the sum of the lead times.

HYPOTHETICAL EXAMPLE, A CUSTOMER REFUND IN AN OFFICE

Process time	80 minutes over five steps
Lead time	4 working days, 1,920 minutes
Activity ratio	80 / 1,920 = 4.2%

Faster work would save minutes; the days are lost in the queues. The figures are invented for illustration.

Sources: Karen Martin & Mike Osterling, McGraw-Hill Education, 2014 (via Karen Martin, “Value Stream Mapping: Lead Time” (author's blog, 2014)); Karen Martin & Associates, Institute of Industrial Engineers, 2009; Lean Enterprise Institute, 2014

The times, the formula and the two-hour case follow Martin (2014); %C&A follows her 2009 slides. The LEI lexicon likewise calls processing time a small fraction of lead time. The example is the editors'.

TOOL OF THE ISSUE

A first map on paper

One flow, a pencil and a walk along it. Start where the customer receives the work and go back step by step; write down what you see and do the timing yourself.

01 The flow which product or request, for which customer, how many a day

02 Steps one box per step, from the customer back; process time under each

03 Waiting between the steps: how many wait, for how long

04 Information who tells each step what to do next, and how

05 The timeline total lead time, total process time, activity ratio

06 One change for the future state where the flow should not wait; who leads it, by when

Sources: Mike Rother & John Shook, Lean Enterprise Institute, 1999 (via Lean Enterprise Institute, introduction and Part I excerpts); Karen Martin & Mike Osterling, McGraw-Hill Education, 2014 (via Karen Martin, “Value Stream Mapping: Lead Time” (author's blog, 2014))

Our proposal, based on Rother & Shook and Martin & Osterling.

Open the tool: Delay Analyzer stivencatalyst.com/tools/delay-analyzer/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Learning to See
Mike Rother & John Shook, Lean Enterprise Institute, 1999 · via Lean Enterprise Institute, introduction and Part I excerpts
https://www.lean.org/wp-content/uploads/2021/01/Learning-to-See-part1.pdf

Lean Lexicon: value-stream mapping; production lead time, processing time, value-creating time
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/value-stream-mapping/

Seeing the Whole Value Stream
Dan Jones & Jim Womack, Lean Enterprise Institute, 2011 · via Lean Enterprise Institute, introduction excerpt
https://www.lean.org/wp-content/uploads/2021/01/seeing-the-whole-introduction.pdf

Applying Value Stream Mapping to Improve the Delivery of Patient Care in the Oncology Day Hospital
Pilar I. Vidal-Carreras, Julio J. Garcia-Sabater & Juan A. Marin-Garcia, International Journal of Environmental Research and Public Health 19(7), 2022
https://doi.org/10.3390/ijerph19074265

The Role of Value Stream Mapping in Healthcare Services: A Scoping Review
Juan A. Marin-Garcia, Pilar I. Vidal-Carreras & Julio J. Garcia-Sabater, International Journal of Environmental Research and Public Health 18(3), 2021
https://doi.org/10.3390/ijerph18030951

A cross-sector review on the use of value stream mapping
Wenchi Shou, Jun Wang, Peng Wu, Xiangyu Wang & Heap-Yih Chong, International Journal of Production Research 55(13), 2017 · via Abstract at Semantic Scholar and Western Sydney University
https://doi.org/10.1080/00207543.2017.1311031

Value Stream Mapping: How to Visualize Work and Align Leadership for Organizational Transformation
Karen Martin & Mike Osterling, McGraw-Hill Education, 2014 · via Karen Martin, "Value Stream Mapping: Lead Time" (author's blog, 2014)
https://www.goodreads.com/author_blog_posts/6062479-value-stream-mapping-lead-time

Metrics-Based Process Mapping (webinar slides)
Karen Martin & Associates, Institute of Industrial Engineers, 2009
https://www.iise.org/uploadedfiles/Webcasts/2009%20Webinar%20Metrics-Based%20Process%20Mapping.pdf

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked "via". The editors' interpretation is marked "Our reading". Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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